BACTOVISION AI

AUTOMATED CLINICAL MICROBIOLOGY REPORT

Sample ID: #TEST-SESSION
Generated: 2026-10-21 23:40:36
Engine: watershed_hybrid
Calibration: 10.0 px/µm

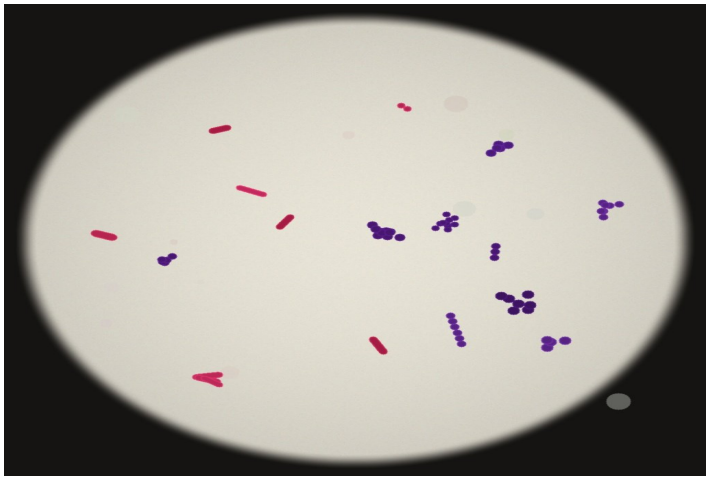
PRIMARY DIAGNOSTIC FINDING: Microscopic automated analysis identified **32** bacterial entities with predominant **Gram-Positive** profile and **Bacilli** morphology.

1. Microscopic Quantification & Gram-Staining Ratios

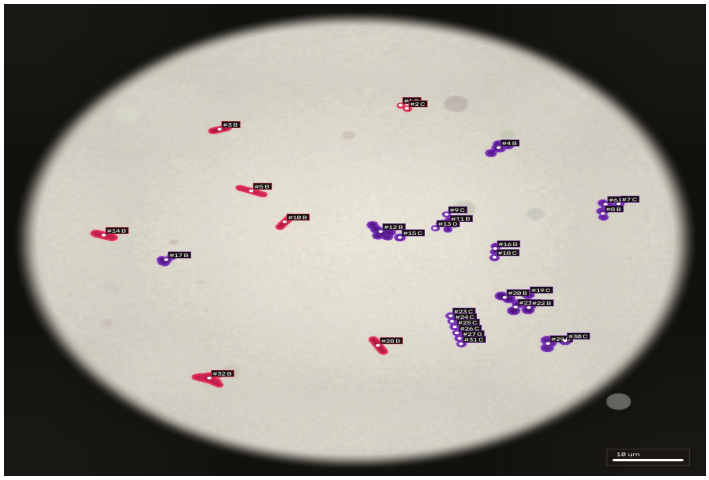
Metric	Count	Percentage	Morphometric Standard	Calculated Value
Gram-Positive	24	75.0%	Mean Cell Area	2.88 µm² (± 2.09)
Gram-Negative	8	25.0%	Mean Circularity	0.688
Indeterminate	0	0.0%	Mean Aspect Ratio	1.76
Total Entities	**32**	**100.0%**	**Cocci / Bacilli Breakdown**	**13 Cocci | 17 Bacilli**

2. Microscopic Field & Instance Segmentation Imagery

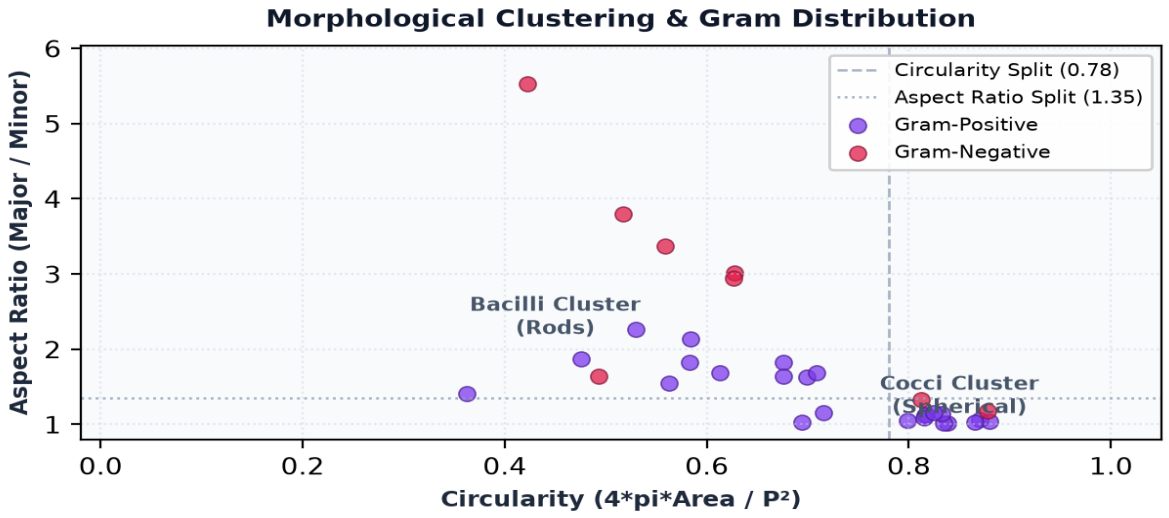
Original Microscope Brightfield



BactoVision AI Annotated Overlay



3. Morphological Clustering: Circularity vs. Aspect Ratio



4. Individual Specimen Morphometric Records (Top 25 Detected Cells)

ID	Gram Status	Morphology	Area (μm²)	Circularity	Aspect Ratio	Confidence
#1	**Negative**	Cocci	0.67	0.878	1.18	99.0%
#2	**Negative**	Cocci	0.90	0.812	1.33	99.0%
#3	**Negative**	Bacilli	3.31	0.627	3.01	99.0%
#4	**Positive**	Bacilli	6.31	0.476	1.87	99.0%
#5	**Negative**	Bacilli	4.08	0.422	5.53	99.0%
#6	**Positive**	Bacilli	2.48	0.699	1.63	99.0%
#7	**Positive**	Cocci	1.06	0.870	1.06	99.0%
#8	**Positive**	Bacilli	2.73	0.583	1.82	99.0%
#9	**Positive**	Cocci	0.80	0.880	1.04	99.0%
#10	**Negative**	Bacilli	3.40	0.559	3.37	99.0%
#11	**Positive**	Bacilli	5.62	0.362	1.41	99.0%
#12	**Positive**	Bacilli	8.54	0.563	1.55	99.0%
#13	**Positive**	Coccobacilli / Other	0.69	0.694	1.03	99.0%
#14	**Negative**	Bacilli	4.60	0.626	2.94	99.0%
#15	**Positive**	Cocci	1.45	0.838	1.02	99.0%
#16	**Positive**	Bacilli	2.21	0.584	2.13	99.0%
#17	**Positive**	Bacilli	3.83	0.613	1.69	99.0%
#18	**Positive**	Cocci	1.08	0.816	1.12	99.0%
#19	**Positive**	Cocci	1.93	0.834	1.01	99.0%
#20	**Positive**	Bacilli	3.83	0.676	1.82	99.0%
#21	**Positive**	Bacilli	3.97	0.530	2.26	99.0%
#22	**Positive**	Bacilli	3.44	0.709	1.68	99.0%
#23	**Positive**	Cocci	1.05	0.799	1.05	99.0%
#24	**Positive**	Cocci	0.97	0.817	1.13	99.0%
#25	**Positive**	Cocci	1.12	0.832	1.14	99.0%

Automated Quality Control:
Image calibration verified with CLAHE normalization. Cell counts validated against Otsu-Watershed segmentation markers.

Verified By: BactoVision AI Medical CV Engine
Classification: Research & Pre-Clinical Diagnostics Assistant